

Functionally Decoupled Visual Intent Learning under Supervisory and Semantic Ambiguity

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Abstract

Visual intention understanding aims to infer the latent purpose conveyed by an image, and is fundamentally more challenging than conventional image classification due to its abstract and subjective nature. A central difficulty of this task lies in two complementary forms of ambiguity: supervisory ambiguity, caused by subjective annotation over plausible intentions, and semantic ambiguity, caused by semantically adjacent and overlapping intents. In this work, we propose FDIL (**F**unctionally **D**ecoupled **I**ntent **L**earning), a framework that treats these two sources of ambiguity through complementary roles. FDIL first introduces a semantic prior built from textual cues to provide structured candidate semantics, and then learns image-aware intent prediction under uncertainty-guided teacher supervision. Concretely, we develop Semantic Local Reranking and Calibration (SLR-C) to form the semantic prior, and Uncertainty-guided Text Distillation (UTD) to transform annotator disagreement into an adaptive training signal. By combining semantic structure with uncertainty-aware learning, FDIL improves both the stability of supervision and the discrimination of semantically similar intents. **On Intentionomy, FDIL improves over the strongest prior method by 6.2% relative average F1 and 28.4% relative hard-subset F1. Extensive experiments demonstrate consistent gains in both hard-sample robustness and overall multi-label prediction performance. Transfers to the larger EMOTIC benchmark and to the Flickr-LDL**

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and Emotion6 label-distribution benchmarks yield smaller gains, while the division of labour between the two modules persists. Overall, FDIL provides a functionally decoupled perspective for visual understanding, and we hope this approach will inspire further research in the field.

Keywords: Visual Intent Recognition, Image Understanding, Multi-Label Image Classification

1. Introduction

With the rapid proliferation of social media, images are increasingly used to share daily experiences, emotions, and personal viewpoints. Visual intention recognition aims to infer the latent purpose conveyed by an image, and has become an important problem in human-centered multimedia analysis [1, 2], with applications ranging from user interest understanding to psychological and behavioral profiling and broader social media content analysis.

Compared with conventional image classification, which assigns images to relatively objective categories such as objects, scenes, or actions, intention recognition must map visual evidence to abstract and subjective intent labels. Such labels are rarely determined by local visual cues alone, and instead emerge from a holistic interpretation of scene context, human activities, object configurations, social interactions, and commonsense knowledge. The main difficulty therefore lies not in visual representation learning, but in bridging concrete image content and abstract intention semantics.

A central challenge of this task is its inherent ambiguity, which manifests in two distinct forms, as shown in Figure 1(a). The first is semantic ambiguity, since intent labels are often semantically adjacent, partially overlapping, or hierarchically correlated, so different labels may correspond to visually similar patterns while the same intent is expressed through diverse appearances, giving high inter-class similarity together with substantial intra-class variation. The second is supervisory ambiguity, since intention recognition is intrinsically subjective and different audiences may infer different yet plausible intentions from the same image, so the supervisory signal itself becomes unstable on samples with severe disagreement, and optimizing directly against hard labels

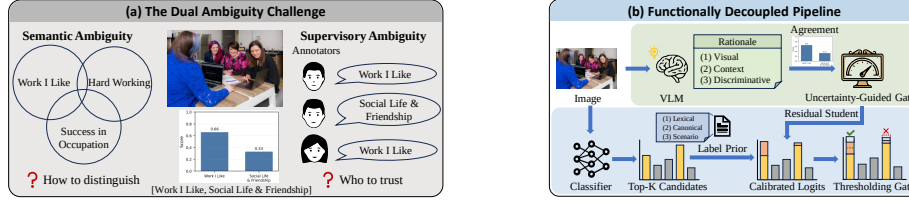


Figure 1: **Functionally decoupled ambiguity modeling in FDIL.** (a) Visual intent recognition suffers from semantic ambiguity among overlapping intents and supervisory ambiguity from annotator disagreement. (b) FDIL separates semantic refinement via Semantic Local Reranking and Calibration (SLR-C) from uncertainty-aware semantic supervision via Uncertainty-guided Text Distillation (UTD).

forces the model to fit oversimplified targets and harms the robustness of the learned decision boundaries. Recent studies increasingly argue that such disagreement should be treated not as annotation noise but as meaningful uncertainty reflecting the structure of subjective interpretation itself [3, 4].

Existing studies have attempted to alleviate these difficulties from different perspectives. Prototype-based methods, *e.g.*, PIP-Net [5], introduce representative anchors to reduce feature-space confusion. Structure-aware methods, including CPAD [6] and HLEG [7], exploit hierarchical label relationships to enhance multi-level semantic modeling and strengthen supervision. From the perspective of label quality, LabCR [8] explicitly models label shifting and label blemish to better cope with imperfect annotation distributions. More recently, knowledge-enhanced approaches have recognized that visual evidence alone is often insufficient for subjective intention understanding. IntCLIP [9] incorporates the vision-language alignment knowledge of CLIP into the task, **HVU-CLIP [10] refines intent semantics iteratively and couples them with bidirectional visual-semantic interaction.** IntentMLM [11] further leverages multimodal large models, caption generation, and retrieval to perform more holistic intention perception, **while CoT4Intent [12] attributes the residual errors of MLLM-based intent classification to hallucination and suppresses it with structured chain-of-thought reasoning.** This line of work reflects a broader shift toward large vision-language and multimodal large language models (MLLMs) as sources of external semantic knowledge for subjective image understanding. However, our zero-shot experiments show

that prompting a general-purpose MLLM directly remains below a task-trained model on this benchmark, since such models supply useful intent priors but not the dataset-specific calibration and uncertainty-aware supervision this task requires. Our purpose is therefore not to replace these models, but to convert their semantic knowledge into a structured prior and an uncertainty-gated training signal that explicitly target the two ambiguities below.

Although these methods have yielded encouraging progress, they typically treat one perspective of ambiguity as dominant. In our view, the core difficulty of subjective visual intention recognition is not any single ambiguity alone, but the coexistence of supervisory ambiguity and semantic ambiguity, which interfere with different functional components of the system. The former primarily corrupts supervision by compressing diverse subjective interpretations into rigid targets, whereas the latter primarily distorts local decision geometry by making neighboring labels difficult to separate. Treating these two failure modes within one mechanism may therefore be suboptimal, as it is difficult to simultaneously achieve robust learning on disagreement-prone samples and precise discrimination among closely related intents.

Motivated by this observation, we propose **FDIL (Functionally Decoupled Intent Learning)**, a functionally decoupled framework for subjective intent recognition, as illustrated in Figure 1(b), where fixed semantic priors provide structured candidate semantics, uncertainty-aware text teachers stabilize ambiguous supervision, and a residual student learns image-aware intent prediction on top of the semantic prior. Ambiguity handling is decoupled into two complementary functions. Semantic Local Reranking and Calibration (SLR-C) provides ambiguity-aware semantic refinement, where heterogeneous lexical, canonical, and scenario priors define a semantic prior supplying structured candidate semantics. Uncertainty-guided Text Distillation (UTD) provides ambiguity-aware semantic supervision, where a text-only teacher distilled from rationale embeddings regularizes the visual model according to annotator agreement, improving robustness under subjective supervision and supporting image-aware prediction on top of the semantic prior. Together they improve robustness and discriminability without high architectural complexity.

Extensive experiments on the Intentionomy dataset show that FDIL outperforms

baselines and recent state-of-the-art methods in both hard-sample robustness and overall multi-label prediction quality. We further transfer FDIL to the larger EMOTIC benchmark and to the Flickr-LDL and Emotion6 label-distribution benchmarks, where the division of labour survives on a larger transfer benchmark and under a different prediction space.

The main contributions of this work are summarized as follows:

- We reformulate visual intention recognition as a *functionally decoupled* ambiguity problem, treating supervisory ambiguity and semantic ambiguity as two separately justified functional roles rather than a single undifferentiated mechanism. This decomposition outperforms a strong unified alternative while keeping each function individually interpretable, separately measurable, and independently deployable.
- We propose Semantic Local Reranking and Calibration (SLR-C) together with residual semantic refinement, where heterogeneous textual priors first define a semantic prior with structured candidate semantics and a lightweight residual student further adapts prediction to image evidence.
- We propose Uncertainty-guided Text Distillation (UTD), which leverages annotator disagreement as an uncertainty signal to adaptively introduce semantic guidance, improving robustness under ambiguous supervision.

2. Related Work

Early visual intent studies target specific domains such as persuasive or political imagery [13, 14, 15], while the Intentionomy benchmark [1] enables general-purpose intent understanding in unconstrained social media. Recent Intentionomy methods address the task from complementary angles. PIP-Net [5] uses prototype-based representations for interpretable intent perception, CPAD [6] and HLEG [7] exploit cross-modality or hierarchical label structures, LabCR [8] models label calibration and relation preservation, and IntCLIP [9] and IntentMLM [11] introduce CLIP/MLLM-based semantic knowledge. More recently, HVU-CLIP [10] extends IntCLIP toward

holistic visual understanding through iterative semantic label refinement and bidirectional visual-semantic interaction, while CoT4Intent [12] investigates hallucination in MLLM-based intention classification and mitigates it through structured chain-of-thought reasoning. Robust-loss methods such as Focal Loss [16], OHEM [17], collaborative refining [18], and ASL [19] reduce the effect of difficult, noisy, or imbalanced samples by reweighting or selective optimization, treating ambiguity as an optimization issue rather than a subjective semantic signal. Disagreement-aware learning instead preserves label diversity. Peterson et al. [3] train on human label distributions, Davani et al. [4] model annotator-specific labels, DisCo [20] jointly models item- and annotator-level distributions, and CrowdOpinion [21] recovers minority opinions. Label distribution learning (LDL) takes the complementary route of predicting the full annotation distribution, where LRR [22] and DPA [23] preserve label ranking relations, δ -LDL [24] relaxes exact distribution recovery to approximate correctness, and LDL-DVS [25] keeps the divisiveness of conflicting labels consistent. These objectives regularize the output distribution but leave the semantics of the label space and the reliability of each supervision signal untouched. In contrast, FDIL treats annotator disagreement as an uncertainty signal and regulates semantic guidance according to supervision ambiguity.

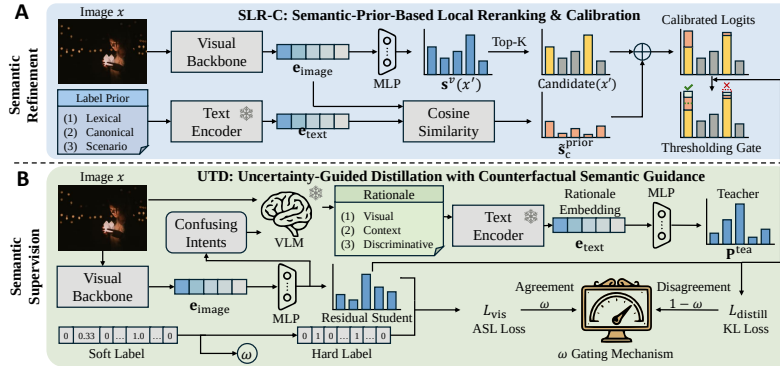


Figure 2: Overview of FDIL. SLR-C constructs a textual semantic prior and performs local reranking plus residual refinement. UTD uses a rationale-based text teacher and an agreement-aware gate to stabilize learning under ambiguous supervision.

Table 1: **Key notation used in the method.**

Symbol	Meaning	Symbol	Meaning
x	input image	$s_c^{\text{prior}}, \bar{s}_c^{\text{prior}}$	raw / normalized prior score
$\mathcal{Y}, C$	label space, no. of classes	$\mathbf{e}_c$	class-level text embedding
$\mathbf{y}^{\text{soft}}, \mathbf{y}^{\text{hard}}$	soft / binarized labels	$C_K(x), K$	Top- K candidate set, its size
$\omega \in (0, 1]$	sample-level agreement	α	prior strength in reranking
$f_{\text{img}} = \phi$	frozen visual encoder	$\tilde{s}_c$	locally reranked logits
$\mathbf{s}^b(B)$	frozen base visual logits	s_c^{full}	final refined score for class c
h_{res}	residual student MLP	t_c	class-wise threshold (validation)
$T(x)$	rationale text for image x	$\mathbf{p}^{\text{tea}}, \tau$	teacher distribution, temperature
$\mathbf{s}^v$	student logits under KD	$g = 1 - \omega$	teacher-branch gate weight
$\mathcal{L}_{\text{vis}}, \mathcal{L}_{\text{distill}}$	classification / KD loss	$\lambda, \mathcal{L}_{\text{total}}$	KD weight, total loss

3. Method

3.1. Overview

To address the dual challenges of supervisory and semantic ambiguity, we propose **FDIL**, a functionally decoupled framework where fixed semantic priors provide structured candidate semantics, uncertainty-aware text teachers stabilize ambiguous supervision, and a residual student learns image-aware intent prediction on top of the semantic prior. Instead of forcing one unified mechanism to handle both ambiguities, we assign them to two complementary functions.

FDIL consists of two key components, as shown in Figure 2.

- **Semantic Local Reranking and Calibration (SLR-C)**, which constructs a semantic prior from heterogeneous textual cues and supports image-aware refinement through a residual student.
- **Uncertainty-guided Text Distillation (UTD)**, which introduces adaptive semantic guidance to improve robustness under ambiguous supervision.

3.2. Problem Formulation

Given an input image x , the goal is to predict a set of intent labels from a predefined label space $\mathcal{Y} = \{1, \dots, C\}$, where C is the total number of intent categories. Due to the subjective nature of visual intention recognition, annotations are provided as soft label vectors $\mathbf{y}^{\text{soft}} \in \{0, 0.33, 0.66, 1\}^C$, reflecting annotator agreement.

Existing approaches typically follow either hard binarization or soft regression, both limited under ambiguous supervision, so we adopt a **decoupled utilization** of annotation signals. Specifically, we derive binarized labels $\mathbf{y}^{\text{hard}}$ for classification supervision, while extracting an image-level agreement score $\omega \in (0, 1]$ from $\mathbf{y}^{\text{soft}}$. Here, ω reflects annotator consensus, where higher values indicate more reliable supervision. The binarized labels are computed as

$$y_c^{\text{hard}} = \begin{cases} 1, & \text{if } y_c^{\text{soft}} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

In practice, an image may contain multiple positive intents with different agreement levels. We therefore adopt a conservative pooling rule and define the sample-level agreement as the minimum soft agreement among all positive labels.

$$\omega = \min_{c: y_c^{\text{hard}}=1} y_c^{\text{soft}}. \quad (2)$$

This choice lets the hardest positive label determine how much the sample should trust direct supervision.

3.3. Semantic Local Reranking and Calibration (SLR-C)

Semantic ambiguity mainly manifests as local confusion among visually plausible but semantically adjacent intents. A candidate-recall diagnostic shows that the baseline Top-10 candidates already cover 83.9% of positive labels and that 67.8% of its false negatives remain recoverable within that set, so many errors arise from ranking confusable intents rather than from failing to retrieve them. We therefore build a semantic refinement module supplying structured candidate semantics before turning to ambiguity-aware supervision.

3.3.1. Semantic Prior Construction

Intent semantics are multi-granular and abstract, so a single textual representation is insufficient. We therefore build a heterogeneous semantic prior per intent from three complementary textual sources: lexical intent phrases giving concise name cues, canonical intent descriptions giving semantic definitions, and descriptive visual scenarios grounding abstract intents in plausible visual situations.

Example. For the intent *SocialLifeFriendship*, the lexical phrase is “*Social Life and Friendship*”, the canonical description is “*Being part of a social group. Having people to do things with. Having close friends, others to rely on. Making friends, drawing others near.*”, and the scenario source expands it into distinct visual realizations such as friends hiking together, close friends talking on a couch, a group dinner, or volunteers gardening together.

The scenarios are not hand-picked but produced by a fixed generation protocol applied uniformly to every intent class (template released with our artifacts), under which the vision-language model enumerates visually distinct realizations along five axes (subjects/objects, actions, settings, atmosphere, and photographic style), with at least three settings per intent so as to span intra-class variation rather than a single canonical view. Outside the generated scenarios the prior degrades gracefully toward the lexical and canonical sources, the base logits, and the residual student.

Given an image x , we score each intent by the CLIP-scaled cosine similarity between the image feature $f_{\text{img}}(x)$ and the text embedding $\mathbf{e}_c$ of class c .

$$s_c^{\text{prior}} = \text{sim}(f_{\text{img}}(x), \mathbf{e}_c) = \gamma \cdot \frac{f_{\text{img}}(x)^\top \mathbf{e}_c}{\|f_{\text{img}}(x)\|_2 \|\mathbf{e}_c\|_2}, \quad (3)$$

where γ is the CLIP logit scale. Each source $r \in \{\text{lex}, \text{can}, \text{scen}\}$ is normalized independently across the C classes and the three are averaged into the prior.

$$\tilde{s}_c^r = \frac{s_c^r - \mu^r(x)}{\sigma^r(x) + \epsilon}, \quad \tilde{s}_c^{\text{prior}} = \frac{1}{3} \sum_r \tilde{s}_c^r, \quad (4)$$

where $\mu^r(x)$ and $\sigma^r(x)$ are the per-sample mean and standard deviation over the C intent classes and ϵ ensures numerical stability.

3.3.2. Semantic Local Reranking

In the refinement module, we start from base visual logits $\mathbf{s}^b$ produced by a frozen visual student. We first select a compact candidate set as

$$C_K(x) = \text{TopK}(\mathbf{s}^b, K). \quad (5)$$

The base logits are then locally adjusted with the normalized prior.

$$\tilde{s}_c = \begin{cases} s_c^b + \alpha \cdot \tilde{s}_c^{\text{prior}}, & c \in C_K(x), \\ s_c^b, & \text{otherwise.} \end{cases} \quad (6)$$

Here α controls the strength of semantic reranking and is fixed to 0.3. This localized reranking narrows the candidate set and sharpens the local score geometry without adding a trainable prior module.

3.3.3. Residual Refinement on the Semantic Prior

Fixed reranking alone is insufficient, since the heterogeneous prior may still fall short of image-specific evidence, so we train a lightweight residual student on top of the semantic prior.

$$s_c^{\text{full}} = \tilde{s}_c + h_{\text{res}}(f_{\text{img}}(x))_c, \quad (7)$$

where $h_{\text{res}}(\cdot)$ is a residual MLP predicting class-wise residual intent scores from the image feature. The SLR-C logits $\tilde{s}_c$ are computed and frozen first, and the residual module is optimized on top of them. In the full model, it is additionally regularized by the uncertainty-aware text teacher introduced next, so SLR-C supplies the structured candidate semantics and UTD the ambiguity-aware supervision.

3.3.4. Class-wise Calibration

Following Guo et al. [26], to account for class imbalance and varying score distributions, we adopt class-wise thresholds.

$$\hat{y}_c = \mathbb{I}[\sigma(s_c^{\text{full}}) > t_c], \quad (8)$$

where $\hat{y}_c$ is the predicted label for class c , $\mathbb{I}[\cdot]$ is the indicator function, $\sigma(\cdot)$ is the sigmoid function, not to be confused with the per-source standard deviation $\sigma^r(x)$ above, and t_c is determined on the validation set by maximizing the class-wise validation F1 score.

$$t_c^* = \arg \max_{t \in \mathcal{T}} \text{F1}_c^{\text{val}}(t), \quad (9)$$

where $\mathcal{T}$ is a fixed threshold grid, so each class adapts its decision boundary to its own score distribution and frequency instead of sharing a global threshold.

3.4. Uncertainty-guided Text Distillation (UTD)

Supervisory ambiguity arises when multiple plausible intents coexist for an image, where direct optimization against hard labels leads to overconfident predictions and

unstable decision boundaries. We therefore introduce semantic guidance from text space and adapt its influence to the uncertainty of supervision.

3.4.1. Text-derived Teacher

For each training image, we construct structured textual descriptions capturing semantic cues relevant to intention understanding, generated offline by a vision-language model from both the image content and intent semantics.

To enhance discrimination under semantic ambiguity, we also select high-confidence negative intents predicted by the current model as confusion candidates and use them to guide generation, which encourages the text to emphasize subtle distinctions among semantically adjacent intents.

The generated text aims to summarize three aspects. (1) observable visual elements, (2) contextual interpretation beyond direct perception, and (3) discriminative cues for distinguishing confusing intents.

The resulting text $T(x)$ is encoded into a dense embedding space using a frozen text encoder.

$$\mathbf{e}_{\text{text}} = f_{\text{text}}(T(x)). \quad (10)$$

A lightweight classifier then maps the embedding into a multi-label teacher distribution.

$$\mathbf{p}^{\text{tea}} = \sigma(g_{\text{tea}}(\mathbf{e}_{\text{text}})/\tau), \quad (11)$$

where g_{tea} is a lightweight classifier head and τ is the temperature parameter.

These contrastive signals improve the teacher’s ability to separate similar intents while remaining independent of the semantic prior, and in the full pipeline the same teacher also regularizes the residual refinement module.

3.4.2. Uncertainty-guided Distillation

The visual backbone produces logits $\mathbf{s}^v$ for image x . We adopt the Asymmetric Loss [19] for classification.

$$\mathcal{L}_{\text{vis}} = \frac{1}{C} \sum_{c=1}^C \ell_{\text{ASL}}(s_c^v, y_c^{\text{hard}}). \quad (12)$$

To incorporate semantic guidance, we introduce an uncertainty-guided Bernoulli distillation loss.

$$\mathcal{L}_{\text{distill}} = \sum_{c=1}^C \text{KL}(\text{Bern}(p_c^{\text{tea}}) \parallel \text{Bern}(\sigma(s_c^v/\tau))), \quad (13)$$

which is applied independently to each label. We use Bernoulli rather than softmax distillation because intent recognition is multi-label, so each intent is an independent Bernoulli variable rather than a mutually exclusive class on a simplex, which preserves per-class uncertainty and avoids artificial competition among labels.

Rather than mixing the two objectives with a fixed global coefficient, we use a gate derived from the agreement score. For the current sample, the teacher branch weight is defined as $g = 1 - \omega$, so low-agreement samples receive stronger semantic guidance, while high-agreement samples remain dominated by direct visual supervision.

Using g , the final loss is defined as

$$\mathcal{L}_{\text{total}} = (1 - g)\mathcal{L}_{\text{vis}} + g\mathcal{L}_{\text{distill}} = \omega\mathcal{L}_{\text{vis}} + \lambda(1 - \omega)\mathcal{L}_{\text{distill}}, \quad (14)$$

where λ is the distillation loss weight.

The model therefore relies on visual supervision when annotations are reliable and on semantic guidance when supervision is uncertain, so UTD stabilizes ambiguous supervision through the text teacher rather than by modifying the semantic prior.

3.5. Functionally Decoupled Ambiguity Modeling

Three sources of residual uncertainty. Let X denote the image, $Z = \phi(X)$ its deterministic frozen visual representation, $Y_c \in \{0, 1\}$ a random human judgement for intent c , and $B = s^b(Z)$ the baseline decision scores. Since B is a function of Z and Z of X , chaining $I(Y_c; Z \mid B) = H(Y_c \mid B) - H(Y_c \mid Z)$ with $I(Y_c; X \mid Z) = H(Y_c \mid Z) - H(Y_c \mid X)$ yields the exact identity

$$H(Y_c \mid B) = \underbrace{H(Y_c \mid X)}_{\text{irreducible supervisory}} + \underbrace{I(Y_c; X \mid Z)}_{\text{discarded by } \phi} + \underbrace{I(Y_c; Z \mid B)}_{\text{unused by the readout}}, \quad (15)$$

where both conditional mutual informations use $H(Y_c \mid X, Z) = H(Y_c \mid X)$ and $H(Y_c \mid Z, B) = H(Y_c \mid Z)$ because Z and B are deterministic functions of X and Z . The first term is disagreement persisting given the complete image; being a functional of the

annotation distribution $\Pr(Y_c \mid X)$ alone, no choice of ϕ and no predictor reading Z can reduce it. The second is label-relevant information that X carries but the frozen Z discards, a functional of the representation map that is unrecoverable downstream by the data-processing inequality. The third is information already carried by Z that the baseline decision rule leaves unexploited, a functional of the readout. Since the backbone stays frozen, FDIL does not act on the second term; SLR-C acts on the third, while UTD does not reduce the first but changes how strongly supervision drawn from it is trusted. The three terms are functionals of different objects, which motivates treating them with distinct mechanisms; whether that separation pays off empirically is tested against a unified control in Sec. 4.3.2.

SLR-C recovers information the readout leaves unused. Let $S = \tilde{s}^{\text{prior}}(X) = \psi(Z)$ be the text-side prior scores, which are similarities between the same frozen image feature and class-level text embeddings and are therefore themselves a function of Z . Conditioning additionally on the prior gives the identity

$$H(Y_c \mid B, S) = H(Y_c \mid B) - I(Y_c; S \mid B), \quad (16)$$

so residual uncertainty falls by $I(Y_c; S \mid B)$, the label-relevant information in the prior that the baseline scores do not already carry, which is zero precisely when S is redundant given B . Since $S = \psi(Z)$, this gain is bounded by the third term of (15), $I(Y_c; S \mid B) \leq I(Y_c; Z \mid B)$, so SLR-C recovers part of what the readout leaves unused and cannot recover what ϕ discarded. It injects under-used text-side information rather than providing a second view of the image, and $\tilde{s}_c = s_c^b + \alpha \tilde{s}_c^{\text{prior}}$ is the minimal way to make the decision rule a function of (B, S) rather than of B alone.

Top- K reranking as local hypothesis screening. For a positive intent c^+ and a confusing intent c^- in $C_K(x)$, let $m = s_{c^+}^b - s_{c^-}^b$ and $\Delta_{\text{prior}} = \tilde{s}_{c^+}^{\text{prior}} - \tilde{s}_{c^-}^{\text{prior}}$, and the reranking rule gives

$$m' - m = \alpha \Delta_{\text{prior}}, \quad (17)$$

so the margin grows whenever the prior ranks c^+ above c^- , a reversed pair is corrected when $\alpha \Delta_{\text{prior}} \geq -m > 0$, and scores outside $C_K(x)$ are untouched. Restricting the prior to $C_K(x)$ is thus local hypothesis screening, in the sense of variable screening for high-dimensional selection: the semantic channel acts only on hypotheses the visual

evidence already deems plausible, and K balances coverage against distractor noise. SLR-C therefore does not globally rescore the label space but corrects confusion among high-probability candidates.

UTD as reliability-weighted privileged information. The rationale teacher reads a textual description $T(x)$ generated with access to intent semantics and model-specific confusion candidates, which exists at training time only. UTD therefore uses privileged information [27], realized through distillation [28], and must decide how much to trust that privileged channel. Its role is not to reduce the irreducible term of (15), which no learner can do, but to limit the error of estimating a subjective target from finitely many annotators and to avoid overfitting hard targets that a low-agreement sample cannot support. In a per-sample scalar view of one label, the annotated target is $y_v = y + \varepsilon_v$ with $\mathbb{E}\varepsilon_v = 0$ and $\text{Var}(\varepsilon_v) = \sigma_v^2(\omega)$, decreasing in the agreement score ω since higher consensus means less annotation noise, while the teacher gives $y_t = y + b_t + \varepsilon_t$ with bias b_t and independent noise of variance σ_t^2 . For $\hat{y}(g) = (1 - g)y_v + g y_t$ we have $\text{MSE}(g) = (1 - g)^2\sigma_v^2(\omega) + g^2(\sigma_t^2 + b_t^2)$, minimized at

$$g^* = \frac{\sigma_v^2(\omega)}{\sigma_v^2(\omega) + \sigma_t^2 + b_t^2}, \quad (18)$$

which increases with supervision noise, decreases with teacher noise and bias, and is monotonically decreasing in ω . Our gate $g = 1 - \omega$ is a monotone, parameter-free approximation of (18). It keeps the correct direction without estimating σ_t^2 or b_t^2 , and reaches $g = 0$ at $\omega = 1$, where the observed supervision is maximally reliable, so the privileged channel can contribute little beyond its own bias. The two coincide exactly only if unanimous annotators imply $\sigma_v^2(1) = 0$, which we do not assume, since unanimity among a finite panel need not mean noiseless population supervision. The teacher contribution to the student-logit gradient is then

$$\frac{\partial \lambda(1 - \omega)\mathcal{L}_{\text{distill}}}{\partial s_c^v} = \frac{\lambda(1 - \omega)}{\tau}(p_c - q_c), \quad (19)$$

with $p_c = \sigma(s_c^v/\tau)$ and $q_c = p_c^{\text{lea}}$, so guidance is routed in proportion to annotator disagreement.

In summary, SLR-C addresses readout insufficiency in the decision space, while UTD addresses unreliable supervision through uncertainty-weighted privileged infor-

mation. Since each module is tied to a different source of uncertainty identified in (15), their gains should be distinctly sensitive to the two ambiguity sources, which we test in Sec. 4.3.2.

4. Experiments

4.1. Experimental Setup

4.1.1. Dataset and Evaluation Metrics

We evaluate our method on the Intentionomy dataset [1], a multi-label benchmark for visual intention recognition in social media scenarios, using the standard split of 12,739 training, 498 validation, and 1,216 test images. Its training set provides soft agreement annotations with values in $\{0, 0.33, 0.66, 1\}$ that reflect annotator consensus and serve as the uncertainty signal ω .

We report Macro, Micro, and Samples F1 and, following recent Intentionomy works, their average Avg.; unless stated otherwise all numbers are percentages on the Intentionomy test split. Following Jia et al. [1], we also break intent classes down by difficulty, measured by how much a standard ResNet-50 model exceeds random performance, and by content dependency into object-dependent (*Object*), context-dependent (*Context*), and *Others*, which depend on both foreground and background. We emphasize the *Hard* score throughout, since it measures the ambiguity-prone categories where semantic overlap and annotator disagreement are most severe.

The cross-dataset study uses three further benchmarks. EMOTIC [29] is a context-based affective benchmark with 23,571 images and 34,320 annotated instances, labelling each annotated person with 26 non-exclusive emotion categories. It is our larger-scale transfer target because it is also multi-label, and we report mAP alongside the three F1 scores on it. Flickr-LDL [30] (11,150 Flickr images, eight emotions) and Emotion6 [31] (1,980 images, seven categories including neutral) are label-distribution benchmarks, for which we report the seven standard label-distribution metrics, which are Chebyshev, Clark, KL divergence (KLD), cosine similarity, Spearman rank correlation, the parameter-free approximate-correctness percentage μ [24], and the divisiveness error DVSE [25], which measures how much conflicting positive and negative

emotion mass a prediction preserves and is therefore the metric closest to the ambiguity we target.

Table 2: Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. Matched-protocol rows (the “(ViT-L/14)” reproductions, the CLIP baseline, and ours) are five-seed means, $\pm$ the 95% t -interval half-width. [†]HVU-CLIP is quoted under its own ViT-B/32 backbone since no code is released; Table 14 gives a backbone-matched comparison.

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [32]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
MultiGuard [33]	NeurIPS 2022	26.87	38.67	40.06	35.20
HiMulConE [34]	CVPR 2022	28.43	41.50	42.15	37.36
SST [35]	AAAI 2022	27.33	42.21	42.62	37.39
DualCoOp [36]	NeurIPS 2022	31.06	44.42	40.40	38.63
Query2Label [37]	-	32.12	44.64	45.15	40.64
Label noise methods					
Cleannet [38]	CVPR 2018	25.89	42.15	40.97	36.34
LS [39]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [40]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
PIP-Net [5]	TMM 2023	32.57	45.94	47.05	41.85
CoT4Intent [12]	-	41.70	53.10	53.82	49.54
HVU-CLIP [10] (ViT-B/32) [†]	TPAMI 2026	42.91	55.22	52.54	50.22
IntentMLM [11]	PR 2026	43.05	54.77	56.75	51.52
IntCLIP (ViT-L/14) [9]	ECCV 2024	44.98 $\pm$ 1.12	54.73 $\pm$ 0.49	54.73 $\pm$ 1.18	51.48 $\pm$ 0.82
HLEG (ViT-L/14) [7]	TAC 2023	47.38 $\pm$ 0.75	54.85 $\pm$ 1.70	55.29 $\pm$ 1.06	52.50 $\pm$ 0.98
LabCR (ViT-L/14) [8]	TIP 2024	47.87 $\pm$ 0.38	56.78 $\pm$ 0.97	57.23 $\pm$ 1.25	53.96 $\pm$ 0.71
CLIP ViT-L/14 [41]	ICML 2021	47.33 $\pm$ 1.15	54.98 $\pm$ 2.90	55.74 $\pm$ 2.64	52.68 $\pm$ 1.96
FDIL (UTD only)	Ours	49.90 $\pm$ 1.18	58.01 $\pm$ 1.03	58.43 $\pm$ 1.13	55.45 $\pm$ 1.02
FDIL	Ours	51.90$\pm$0.95	59.90$\pm$0.62	60.03$\pm$0.94	57.28$\pm$0.75

4.1.2. Implementation Details

Our default backbone is frozen CLIP ViT-L/14 [41], and the visual student is a lightweight residual MLP on top of a fixed SLR-C prior. In SLR-C, priors come from lexical intent phrases, canonical intent descriptions, and Gemini 3.1 Pro-generated scenario descriptions [42] encoded by the frozen CLIP text encoder, their normalized log-ratios averaged before local reranking within Top-10 visual candidates. For UTD, ratio-

nale text is generated offline by Qwen2.5-VL-7B-Instruct [43] and encoded by a frozen bge-large-en-v1.5 text encoder [44], with both prompt templates released with our artifacts. The teacher consumes only text embeddings and uses Asymmetric Loss [19] ($\gamma_- = 2$, $\gamma_+ = 0$, clipping 0.05); the student uses AdamW [45] with learning rate 5×10^{-4} , weight decay 10^{-4} , batch size 256, dropout 0.1, hidden dimensions 1,024 (teacher) and 768 (student), and distillation temperature 2.0. Best-epoch selection and early stopping use validation class-wise Macro F1, and all thresholds are fitted on the validation split and fixed before test evaluation, so test outputs never enter model or threshold selection. UTD introduces *zero extra inference cost*, since the text teacher is discarded at test time.

4.2. Comparative Experiments

4.2.1. Comparison with state-of-the-art methods

Table 2 summarizes the main comparison on the test split. Under the matched protocol FDIL leads the reproduced methods by +3.3 to +5.8 Avg. F1, and paired two-sided t -tests over the five shared seeds are significant after Holm correction on all four F1 metrics against every one of them; against the strongest, LabCR, the margin is +3.32 Avg. F1 ($p_H = 0.002$). UTD alone already improves Avg. F1 at zero extra inference cost, and full FDIL is best on every column.

4.2.2. Zero-shot Multimodal Large Model Comparison

To test whether recent multimodal large language models already solve visual intent recognition by prompting alone, we evaluate them zero-shot on the same label set, prompting each for a probability over the candidate categories. Table 3 shows zero-shot capability rising steadily. The open mid-size models reach 36.67 to 40.15 Avg. F1, GPT-4o 40.50, and the frontier GPT-5.6 Sol 49.04, closing much of the gap. It nevertheless still trails the task-trained CLIP baseline by 3.64 Avg. F1 and FDIL by 8.24, so prompting a general-purpose MLLM does not yet substitute for the dataset-specific calibration and uncertainty-aware supervision this task requires, and costs considerably more at inference.

Table 3: Zero-shot MLLM comparison on the Intentionomy test split. [‡]Published result quoted from IntentionMLM [11].

Model	Macro	Micro	Samples	Avg.
Zero-shot prompting				
Qwen3.5-9B [46]	37.11	36.50	36.40	36.67
Step3-VL-10B [47]	38.21	40.21	36.10	38.17
Qwen3-VL-8B-Instruct [48]	38.51	40.75	39.63	39.63
InternVL3-8B [49]	37.54	41.85	40.50	39.97
Gemma-4-E4B-IT [50]	38.76	42.44	39.25	40.15
GPT-4o [51] [‡]	39.48	40.57	41.45	40.50
GPT-5.6 Sol [52]	47.58	49.64	49.91	49.04
Task-trained reference				
CLIP ViT-L/14 [41]	47.33	54.98	55.74	52.68
FDIL (ours)	51.90	59.90	60.03	57.28

4.2.3. Performance on Difficulty- and Content-Dependent Subsets

Table 4 groups intent subsets by difficulty and content dependency, reproducing the matched-protocol methods of Table 2 under identical features and split. **FDIL’s gains concentrate where ambiguity is highest. On the hard subset it reaches 34.66 F1 against 26.99 for the strongest reproduced competitor, LabCR, and every hard-subset contrast against the reproduced methods is significant over the five shared seeds (+7.67 against LabCR, all Holm-corrected $p \leq 0.004$). The medium subset improves by +2.54 to +5.97 (all $p_H \leq 0.018$), whereas on the easy subset no method separates from another ($p_H = 1.00$), those categories leaving little headroom. On the content split the major gain is in *Others*, which needs holistic reasoning over multiple weak cues rather than one object or scene pattern.**

4.2.4. Interpretability Analysis

Figure 3 projects the decision-score space of four semantically adjacent intents. FDIL gives more compact, better-separated same-intent clusters than the CLIP baseline (silhouette $0.21 \rightarrow 0.29$, computed in the original decision-score space rather than on the two-dimensional embedding), so the semantic prior and rationale teacher pull confusable neighbors apart rather than merely rescaling scores. Per-class F1 changes under the matched class-wise protocol concentrate on hard and abstract classes, including *WorkILike* (+28.5), *PassionAboutSomething* (+17.3), and *SocialLifeFriendship*

Table 4: Intent subsets grouped by difficulty and content dependency, where Avg. is the corresponding subset mean. The lower block reproduces the matched-protocol rows of Table 2 under identical features, split, and thresholding, as five-seed means; the ablation tables quote their own seed protocol.

Method	Difficulty				Content dependency			
	Easy	Medium	Hard	Avg.	Object	Context	Others	Avg.
Random	19.86	7.11	2.81	9.93	7.75	12.53	6.05	8.78
Visual [32]	54.64	24.92	10.71	30.09	25.58	30.16	21.34	25.69
Jia (w/o hashtag) [1]	57.10	25.68	12.72	31.83	28.15	28.62	22.60	26.46
Jia (hashtag) [1]	58.86	26.30	13.11	32.76	29.66	32.48	22.61	28.25
Query2Label [37]	81.59	38.06	14.55	44.73	–	–	–	–
CPAD [6]	69.47	31.50	8.54	36.50	28.81	47.30	24.74	33.62
PIP-Net [5]	70.73	33.63	14.33	39.56	39.20	39.53	27.59	35.44
IntentMLM [11]	76.26	46.87	27.39	50.17	50.13	49.69	39.75	46.52
IntCLIP (ViT-L/14) [9]	80.83	51.70	24.15	52.23	57.30	61.37	38.72	52.46
HLEG (ViT-L/14) [7]	80.98	54.79	26.19	53.99	58.98	60.65	41.71	53.78
LabCR (ViT-L/14) [8]	81.17	55.13	26.99	54.43	60.78	61.00	41.73	54.50
CLIP ViT-L/14 [41]	80.89	53.83	27.52	54.08	60.53	63.32	40.79	54.88
FDIL	80.51	57.67	34.66	57.61	63.00	61.33	46.81	57.05

(+17.2), while only two classes regress.

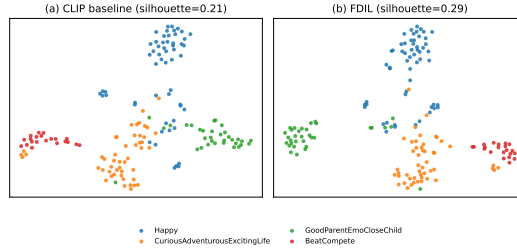


Figure 3: t-SNE of the decision-score space on four semantically adjacent intents. FDIL yields tighter, better-separated same-intent clusters than the CLIP baseline.

4.3. Ablation Studies

4.3.1. Effectiveness of SLR-C and Residual Refinement

Table 5 ablates the two supervision ingredients on top of the SLR-C prior and its residual student. Removing text distillation costs 4.47 Avg. and 5.31 hard-subset F1, and keeping the teacher but removing agreement-aware gating still costs 1.97 and 3.33. The prior and residual student alone are therefore not sufficient, and the gate is

Table 5: Component ablation of FDIL, where *w/o UTD* removes text distillation and *Ungated teacher* removes agreement-aware gating.

Method	Macro	Micro	Samples	Avg.	Hard
w/o UTD	47.90	55.98	55.18	53.02	29.71
Ungated teacher	49.69	57.90	58.98	55.52	31.69
FDIL (full)	52.13	60.06	60.28	57.49	35.02

not a cosmetic reweighting, so prior fusion and uncertainty-aware supervision are best optimized jointly.

4.3.2. Effectiveness and Necessity of Functional Decoupling

Table 6 contrasts decoupled FDIL with three unified controls under identical features, split, and thresholding, where *Unified joint-target KD* averages the SLR-C and rationale-teacher probabilities into one target and trains a single student without residual decomposition or gating. Neither signal is degenerate alone, but the combination rule is decisive.

Table 6: Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	60.10	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL	52.13	60.06	60.28	57.49	35.02

Table 7 examines how the two modules respond to the two sources of ambiguity. Supervisory ambiguity is measured by $1 - \omega$, with $\omega = 1/3$ defining the high-ambiguity stratum, and semantic ambiguity by the negative top-2 margin of the out-of-fold baseline logits, split at the median. Significance uses paired two-sided t -tests on per-sample F1 against the shared baseline, Holm-corrected over the twelve comparisons. The two modules show different sensitivity patterns. SLR-C benefits samples with high semantic ambiguity, by +4.08 and +1.17 F1 under low and high supervisory ambiguity respectively, both significant after correction. UTD has essentially no effect when agreement is high (-0.05 and -0.07), and its only significant gain is where high supervisory and high semantic ambiguity coincide (+0.84, corrected $p = 0.025$). SLR-C

therefore tracks label semantics while UTD emerges only under uncertain supervision, supporting the assignment of the two ambiguity sources to separate mechanisms.

Table 7: 2×2 ambiguity-stratified analysis on cross-fitted Intentonomy train predictions. Cells give Δ per-sample F1 against the shared baseline, and * marks Holm-corrected $p < 0.05$.

		Low supervisory ($\omega \geq 2/3$)		High supervisory ($\omega = 1/3$)	
		Low sem.	High sem.	Low sem.	High sem.
n		1,271	588	5,099	5,781
Base	F1	72.23	41.56	40.54	22.90
	+SLR-C	-0.43	+4.08*	+0.49	+1.17*
Δ F1	+UTD	-0.05	-0.07	+0.03	+0.84*
	Full FDIL	-0.71	+2.18	+0.15	+1.14*

4.3.3. Heterogeneous Priors in SLR-C

Table 8 compares the three SLR-C sources under the reranking stage only: *Lexical* (short class-name phrases), *Canonical* (official Intentonomy descriptions), and *Scenario* (Gemini-generated visual scenarios).

Table 8: Ablation of textual priors in SLR-C (reranking-only, $K=10$). Results use a single seed.

Lexical	Canonical	Scenario	Macro	Micro	Samples	Avg.	Hard
✓			49.76	59.12	58.60	55.83	31.22
	✓		49.80	59.50	58.05	55.78	29.88
		✓	49.35	58.88	57.87	55.37	32.48
✓	✓		51.18	59.39	58.11	56.23	32.92
✓		✓	50.12	59.30	58.36	55.93	32.75
	✓	✓	50.93	59.51	58.18	56.21	34.81
✓	✓	✓	50.90	59.81	58.47	56.40	32.80

No single source dominates, so the choice of prior is a trade-off rather than a strict ordering, while the three-source ensemble is best on the aggregate metrics and carries the broadest semantic coverage. We therefore adopt it, which also gives the strongest hard-subset result under the full pipeline.

4.3.4. Sensitivity to Candidate Size K

Table 9 retrains FDIL under $K \in \{5, 10, 28\}$ with class-wise thresholding. $K = 10$ is best and is our default, $K = 5$ slightly weaker, and $K = 28$ comparable but no better, so a moderate candidate set balances semantic coverage against distractor noise.

Table 9: Sensitivity to candidate size K (class-wise thresholding). Results are means over three seeds.

Top- K	Macro	Micro	Samples	Avg.	Hard
5	51.19	59.03	59.42	56.54	31.60
10	52.13	60.06	60.28	57.49	35.02
28	51.64	59.59	59.71	56.98	34.02

Table 10: Calibration decomposition under matched global/class-wise thresholding. FDIL is mean $\pm$ std over three seeds, and CLIP uses a single seed.

Method	Threshold	Macro	Micro	Samples	Avg.	mAP	Hard
Baseline	Global	46.22	55.05	54.51	51.92	50.67	26.84
Baseline	Class-wise	47.40	56.42	56.41	53.41	50.67	28.45
FDIL	Global	47.70 $\pm$ 1.55	57.56 $\pm$ 0.41	57.43 $\pm$ 0.69	54.23 $\pm$ 0.72	55.10 $\pm$ 0.26	30.00 $\pm$ 3.46
FDIL	Class-wise	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49	55.10 $\pm$ 0.26	35.02 $\pm$ 1.77

4.3.5. Calibration and Threshold-free Evaluation

Table 10 evaluates both models under matched global and class-wise thresholding. FDIL improves Avg. and hard-subset F1 under both, and class-wise thresholding helps both, so calibration is a shared post-processing gain rather than the source of FDIL’s advantage, which the decision-rule-invariant mAP confirms with a +4.4 gain.

Table 11: Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds, and all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
Baseline	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

Table 11 and Figure 4 report threshold-free, rank-based metrics. FDIL improves over the baseline on every metric with a 95% paired-bootstrap interval excluding zero ($p < 0.001$).

4.3.6. The Effectiveness of Rationale

Table 12 ablates the teacher text under gated distillation without SLR-C, comparing a plain Caption against rationales built from visual evidence (Step 1), contextual interpretation (Step 2), and counterfactual disambiguation (Step 3).

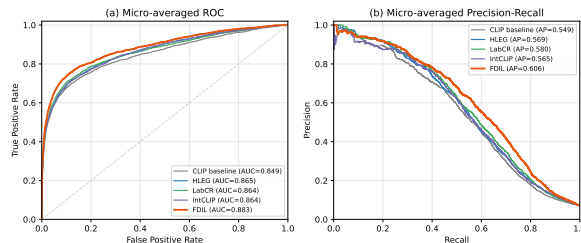


Figure 4: Micro-averaged ROC (a) and precision-recall (b) curves, comparing FDIL against the CLIP baseline and three representative methods (HLEG, LabCR, IntCLIP) reproduced under the matched CLIP ViT-L/14 features, split, and protocol.

Table 12: Ablation of rationale construction for UTD, where Steps 1, 2, and 3 add visual evidence, contextual bridging, and counterfactual disambiguation. Results use a single seed.

Method	Macro	Micro	Samples	Avg.	Hard
Caption	49.64	58.64	58.29	55.52	29.14
Rationale (Step 1)	49.81	56.39	57.65	54.62	29.28
Rationale (Step 1 & 2)	49.69	54.45	55.97	53.37	28.82
Rationale (Step 1 & 2 & 3)	50.31	57.97	59.07	55.78	30.24

Caption supervision is already a strong privileged signal, yet trails the full rationale on Macro, Samples, Avg., and Hard. Adding contextual interpretation (Step 2) slightly degrades the metrics, indicating semantic drift without explicit disambiguation, and the counterfactual Step 3 recovers and surpasses it. Subjective intent supervision therefore needs boundary contraction among confusing alternatives, not only richer context.

4.3.7. Negative Controls on the Text Teacher

Table 13 retrains FDIL with one ingredient corrupted: shuffled rationale (each image paired with another’s), ungated teacher (standard KD), and uniform gate (every sample at the mean agreement), against the supervised residual (w/o UTD) as the lower reference. Shuffling rationales degrades more than removing UTD altogether (−10.40 against −4.47 Avg. F1), so the gain relies on image-specific rationale content rather than arbitrary external text. Flattening or removing the gate degrades less, and the uniform gate even gains 1.13 on the hard subset, but both lose Avg. F1 and every aggregate metric, supporting uncertainty-aware gating.

Table 13: Negative controls on the UTD text teacher, with Δ in points, where negative denotes degradation.

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD	-4.23	-4.08	-5.10	-4.47	-4.93	-5.31
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-2.44	-2.16	-1.30	-1.97	-1.49	-3.33
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

4.4. Backbone Independence

Table 14 compares FDIL with published ResNet101 results, where it remains clearly effective, so the decoupled design is not exclusive to CLIP features. **SLR-C here uses $K=28$ rather than the $K=10$ default, the weaker backbone benefiting from a wider scaffold.** Against HVU-CLIP, FDIL is higher on Macro, Samples, and Avg. F1, while HVU-CLIP is higher on Micro F1, which could be attributed to the concentration on hard categories of FDIL.

Table 14: Comparison of models under ResNet101 on the Intentionomy test split.

Method	Macro	Micro	Samples	Avg.
Baseline	24.71	44.07	44.03	37.60
PIP-Net [5]	30.67	43.44	43.31	39.14
Query2Label [37]	32.12	44.64	45.15	40.64
HLEG [7]	32.77	44.69	45.54	41.00
LabCR [8]	34.63	48.51	48.05	43.73
IntCLIP [9]	38.40	50.54	49.31	46.08
HVU-CLIP [10]	41.24	52.22	50.41	47.96
FDIL (ours)	43.20	50.63	52.02	48.62

4.5. Cross-Dataset Generalization

We transfer FDIL along two axes. EMOTIC is a larger benchmark keeping the multi-label output geometry of Intentionomy, whereas Flickr-LDL and Emotion6 replace binary decisions by a point on the probability simplex and remove threshold selection entirely.

4.5.1. Transfer to a Larger Benchmark

Since annotator agreement is the signal UTD consumes, we evaluate on the multi-annotated pool of 10,613 persons, the only part of EMOTIC that carries it, under a controlled 5×5 cross-validation. Each seed aggregates its complete five-fold out-of-fold performance, so one seed is one paired observation and all tests run on five paired values. Results are therefore internal to this pool rather than comparable with published EMOTIC numbers. Only the candidate size and prior weight of SLR-C are retuned for the new label space ($K=26$, $\alpha=0.09$), and all remaining settings follow Intentionomy.

Table 15: Transfer to EMOTIC: controlled 5×5 cross-validation on the multi-annotated validation+test pool (10,613 persons). Entries are means over five seeds, each aggregating its complete five-fold out-of-fold performance; paired t -tests against the baseline use those five seed-level values. * $p < 0.01$, § $p < 0.05$.

Method	Macro	Micro	Samples	mAP
Baseline	43.34	60.32	60.01	42.57
+ SLR-C	43.42	60.45	60.18	42.70*
+ UTD	43.67	60.88§	60.61§	42.70
FDIL	43.76*	60.99*	60.63*	42.95*

As shown in Table 15, full FDIL improves every metric over the baseline, and each improvement is reliable across the five seeds ($p \leq 0.005$). The division of labour also reappears in a new label space. Gated UTD accounts for the thresholded F1 metrics and leaves mAP unchanged, whereas SLR-C alone moves the F1 metrics only marginally and is the component whose mAP gain is individually reliable. Both mechanisms therefore keep their assigned roles on a larger and semantically more distant benchmark, although the absolute margins are smaller than on Intentionomy.

4.5.2. Transfer to Label Distribution Learning

Under label distribution learning (LDL) the target is a probability distribution and no threshold is selected, testing whether the decoupling generalizes beyond the prediction space it was developed for.

Both functional roles are preserved and only the probability semantics change. Direct supervision becomes categorical cross-entropy against the annotator vote distribution. The image-conditioned residual head is now applied to the distribution logits be-

fore the softmax. UTD replaces the per-class Bernoulli KL with a temperature-scaled categorical KL and reads its gate off the released votes as $\omega = \max_c y_c^{\text{soft}}$. Table 16 reports both datasets.

Table 16: Results on Flickr-LDL and Emotion6, averaged over five seeds. */** indicate significant improvements over the matched baseline at $p < 0.05/p < 0.01$, and † indicates a significant degradation at $p < 0.05$ (paired t -test over five paired runs). Bold and underline mark the best and second-best entry among the baseline and the three FDIL variants.

Method	Cheby.↓	Clark↓	KLD↓	Cos.↑	Spear.↑	μ (%)↑	DVSE↓
<i>Flickr-LDL</i> (2,229 test images)							
LDL-DVS [25]	.20385	2.30341	.34332	.89663	.71085	71.40	.07719
Baseline	.20386	<u>2.30331</u>	<u>.34329</u>	.89664	.71089	71.40	.07723
+ SLR-C	.20308	2.30313	.34061	.89732	.71291	71.58	.07641
+ UTD	.20257	2.31226 †	.34542	.89610	<u>.71369</u>	71.31	.07292*
FDIL	<u>.20272</u>	2.31126 †	.34332	<u>.89672</u>	.71406*	<u>71.45</u>	<u>.07379**</u>
<i>Emotion6</i> (594 test images)							
LDL-DVS [25]	.17939	1.59807	.25322	.89464	.72225	64.20	.07834
Baseline	.17937	1.59790	.25317	.89466	.72227	64.20	.07838
+ SLR-C	<u>.17968</u>	1.59758	<u>.25152</u>	.89548	<u>.72314</u>	64.52	.07811
+ UTD	.18044	<u>1.59491</u>	.25372	.89406	.72563*	64.12	.07907
FDIL	.18017	1.59097*	.25132	<u>.89528</u>	.72194	<u>64.41</u>	<u>.07812</u>

The reproduced LDL objective LDL-DVS [25] stays close to the matched baseline, suggesting that the stronger pretrained representation leaves such regularizers less optimization headroom than the features used in their original evaluations. Our two designed components again act on different quantities. SLR-C mainly improves the distribution-fitting metrics (KLD, cosine similarity, and μ) on both datasets, whereas UTD acts on distribution structure, improving Spearman correlation on both and significantly so on Emotion6 ($p = 0.042$), and significantly reducing DVSE on Flickr-LDL ($p = 0.010$) at the cost of Clark distance. SLR-C thus reallocates probability mass among plausible labels while UTD exploits target uncertainty to shape the predicted structure.

4.6. Efficiency Comparison

Table 17 compares inference efficiency. FDIL (UTD only) costs exactly the baseline, since the text teacher is discarded after training, and full FDIL needs 2.41M pa-

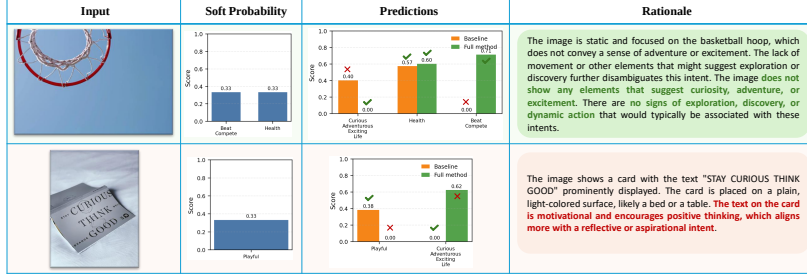


Figure 5: Qualitative case study of baseline and FDIL on representative Intentionomy samples.

rameters, 4.93M FLOPs, and 0.466 ms per image, whereas HLEG introduces much heavier Transformer decoding. IntCLIP uses the fewest parameters of any head measured, though at higher FLOPs than FDIL, and its region-aggregation design runs on the same frozen encoder while trailing FDIL by 5.80 Avg. F1 under the matched protocol of Table 2. HVU-CLIP releases no code and cannot be measured, while IntentMLM and CoT4Intent require MLLM inference and are therefore not included in the measured feature-level comparison. FDIL therefore achieves its gains with minor inference overhead on top of the shared backbone.

Table 17: Feature-level inference cost on pre-extracted CLIP ViT-L/14 features: inference-active parameters, per-image FLOPs, and batch-1 latency on an RTX 4090. The shared frozen encoder is excluded.

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
HLEG [7]	203.70	12 075.9	2.725
IntCLIP [9]	0.02	22.02	0.111
FDIL (UTD only)	1.20	2.40	0.109
FDIL	2.41	4.93	0.466

4.7. Case Study

Figure 5 gives qualitative examples. In the first, FDIL suppresses a plausible but incorrect distractor using counterfactual rationale evidence, consistent with UTD sharpening semantic boundaries. In the second it errs on a 0.33-agreement label yet stays semantically plausible, illustrating that low-agreement samples can admit semantically plausible alternative predictions.

5. Discussion

FDIL rests on the two ambiguities of intent recognition. Keeping them separate makes each function independently justified, measurable, and deployable. It suits offline or high-throughput visual-content analysis, since its external semantic knowledge is prepared offline, the rationale generator and text teacher are discarded after training, and SLR-C adds only lightweight reranking and residual refinement on top of the shared frozen encoder, which remains the dominant inference cost.

The limitations lie in the prior and the visual representation. Offline text generation and prior construction add preparation cost, gains shrink on weaker backbones such as ResNet101, and richer priors are not automatically better unless their interaction with the student is controlled. For a few broad, overlapping categories, the residual student can resolve the candidate set toward a competing neighbor and lose accuracy on the original class. Generalization beyond the intent taxonomy is also limited. Across the transfer studies, the component-level tendencies are broadly consistent, with the semantic prior moving ranking and pointwise distribution fit and the uncertainty-guided teacher moving thresholded F1 and rank correlation, although the overall gains are small and metric-dependent and no single metric improves everywhere. Natural next steps are source-aware prior weighting instead of uniform averaging, stronger students for weaker backbones, and confusion-pair-specific textual evidence.

6. Conclusion

We study subjective visual intention recognition under supervisory and semantic ambiguity and propose FDIL, a functionally decoupled framework where fixed semantic priors provide structured candidate semantics, a residual student supports image-aware prediction on top of that prior, and uncertainty-aware text teachers stabilize ambiguous supervision. Experiments on Intentionomy show consistent improvements over strong visual baselines and prior state-of-the-art methods. Cross-dataset studies on the larger EMOTIC benchmark and on Flickr-LDL and Emotion6 show that the component-level division of labour is broadly preserved, with the semantic prior moving ranking and pointwise distribution fit and the uncertainty-guided teacher moving

thresholded F1 and rank correlation, although the overall gains there are small and metric-dependent. Overall, these results suggest that, given strong visual representations, explicitly modelling supervisory and semantic ambiguity can further improve subjective intent recognition. We hope this decoupled formulation offers a clearer perspective for future research on human-centered visual understanding.

Data and artifact availability

The Intentionomy, EMOTIC, Flickr-LDL, and Emotion6 benchmarks are publicly available. Upon acceptance we will release generated priors, rationales, cached features and logits, prompts, thresholds, seeds, checksums, and training and evaluation scripts.

CRedit authorship contribution statement

Yin Tang: Writing - original draft, Validation, Formal analysis; **Lifeng Fan:** Supervision, Writing - original draft, Conceptualization; **Hongyu Yang:** Writing - original draft, Formal analysis; **Xuan Dong:** Writing - original draft, Formal analysis; **Weixin Li:** Writing - original draft, Supervision, Resources, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT to improve language and readability. The authors then reviewed and edited the content as needed and take full responsibility for the content of the published article.

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